

AI Engineer

Step 5: Learn NLP & Computer Vision

CareerCompass — Detailed Learning Notes

What this step is about

Gain practical exposure to two major AI application areas.

Core Concepts

- NLP: tokenization, normalization, embeddings, text classification, sequence models and transformers.
- Computer vision: image preprocessing, augmentation, CNNs, object detection and evaluation.
- Use pretrained models when they are appropriate and understand inference vs fine-tuning.
- Keep datasets, labels and preprocessing consistent between training and production.

Skills You Should Be Able To Demonstrate

- Build a simple text classifier.
- Preprocess and tokenize text.
- Perform image augmentation.
- Evaluate a vision model with suitable metrics.

Hands-on Project / Practice

- Choose one specialization for a project: sentiment/text classification or image classification/object detection.
- Document data quality, model choice and limitations.

Recommended References

- Hugging Face Documentation — <https://huggingface.co/docs>
- NLP Tutorial with Python & NLTK — <https://www.youtube.com/watch?v=X2vAabgKiuM>
- OpenCV Python Course — https://www.youtube.com/watch?v=P4Z8_qe2Cu0